# Lab Exercise 20- Deploy a Docker Application using AWS ECR and ECS

### 1. Objective

In this lab, you will learn how to:

- Build a container image using **Docker**
- Store the container image in **Amazon Elastic Container Registry**
- Deploy the container on **Amazon Elastic Container Service**
- Access the running application through a public endpoint

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### 2. Prerequisites

Before starting the lab, ensure you have:

- AWS account with appropriate permissions
- Docker installed
- AWS Command Line Interface installed and configured
- Basic understanding of containers
- Internet access

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## 3. Lab Scenario

A development team has built a small web application using containers. Your task as a DevOps engineer is to:

1. Containerize the application using **Docker**
2. Store the container image in **Amazon ECR**
3. Deploy the container to **Amazon ECS**
4. Verify the application is running successfully

---

## 4. Lab Tasks

### Task 1: Create a Docker Application

Create a simple application.

Example

**index.html**

```
Hello from AWS ECS Lab
```

Create a **Dockerfile**

```
FROM nginx

COPY index.html /usr/share/nginx/html/index.html
```

---

### Task 2: Build the Docker Image

Build the image locally using **Docker**.

```
Docker build -t ecs-lab-app .
```

Verify the image.

```
docker images
```

---

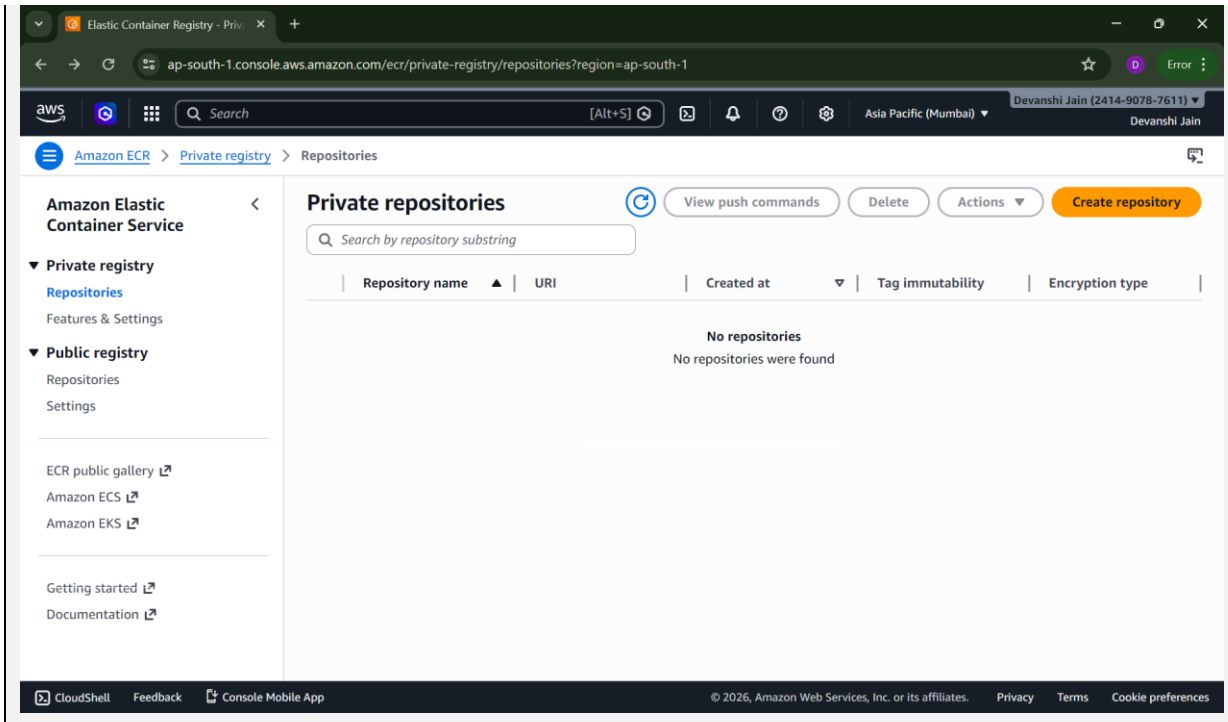
### Task 3: Create an ECR Repository

Create a repository in **Amazon ECR**.

Steps:

1. Open AWS Console
2. Go to **Amazon ECR**
3. Click **Create Repository**
4. Repository name:

```
ecs-lab-repository
```


Elastic Container Registry - Priv
ap-south-1.console.aws.amazon.com/ecr/private-registry/repositories?region=ap-south-1
Search
[Alt+S]
Asia Pacific (Mumbai)
Devanshi Jain (2414-9078-7611)
Devanshi Jain
Amazon ECR > Private registry > Repositories
Amazon Elastic Container Service
Private registry
Repositories
Features & Settings
Public registry
Repositories
Settings
ECR public gallery
Amazon ECS
Amazon EKS
Getting started
Documentation
Private repositories
View push commands
Delete
Actions
Create repository
Search by repository substring
Repository name
URI
Created at
Tag immutability
Encryption type
No repositories
No repositories were found
CloudShell
Feedback
Console Mobile App
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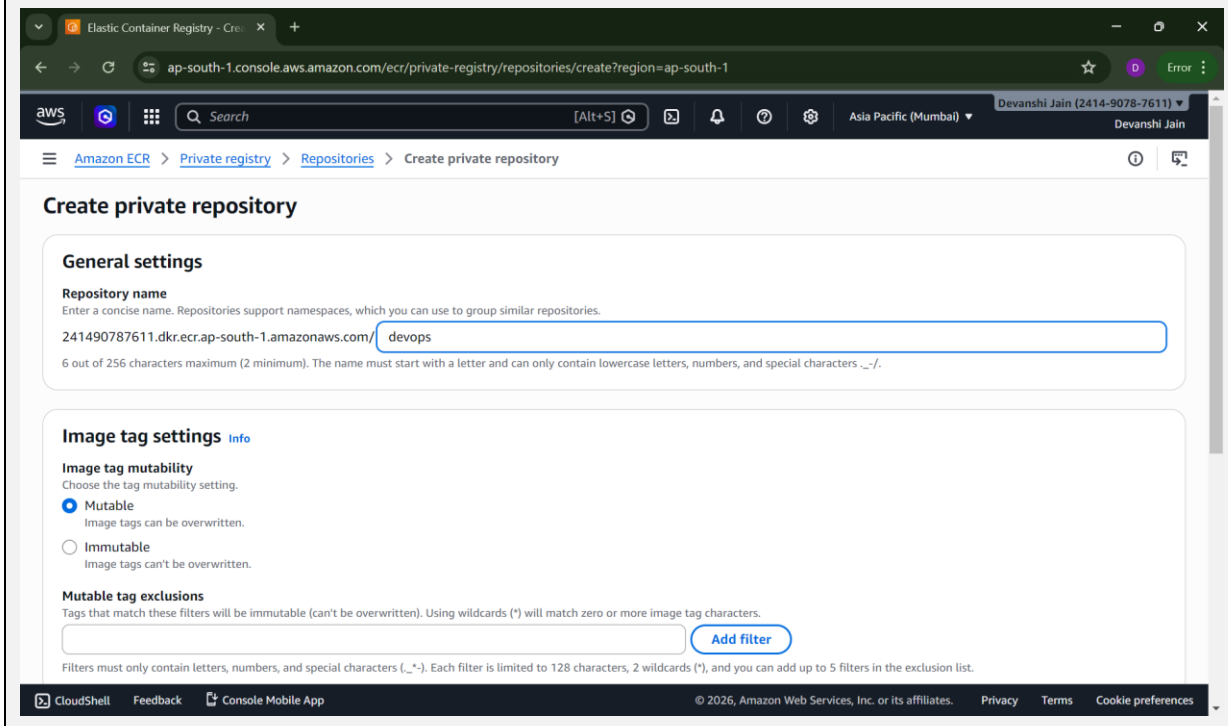
Elastic Container Registry - Cre
ap-south-1.console.aws.amazon.com/ecr/private-registry/repositories/create?region=ap-south-1
Search
[Alt+S]
Asia Pacific (Mumbai)
Devanshi Jain (2414-9078-7611)
Devanshi Jain
Amazon ECR > Private registry > Repositories > Create private repository
Create private repository
General settings
Repository name
Enter a concise name. Repositories support namespaces, which you can use to group similar repositories.
241490787611.dkr.ecr.ap-south-1.amazonaws.com/
devops
6 out of 256 characters maximum (2 minimum). The name must start with a letter and can only contain lowercase letters, numbers, and special characters ._-/.
Image tag settings
Info
Image tag mutability
Choose the tag mutability setting.
Mutable
Image tags can be overwritten.
Immutable
Image tags can't be overwritten.
Mutable tag exclusions
Tags that match these filters will be immutable (can't be overwritten). Using wildcards (*) will match zero or more image tag characters.
Add filter
Filters must only contain letters, numbers, and special characters (._*-). Each filter is limited to 128 characters, 2 wildcards (*), and you can add up to 5 filters in the exclusion list.
CloudShell
Feedback
Console Mobile App
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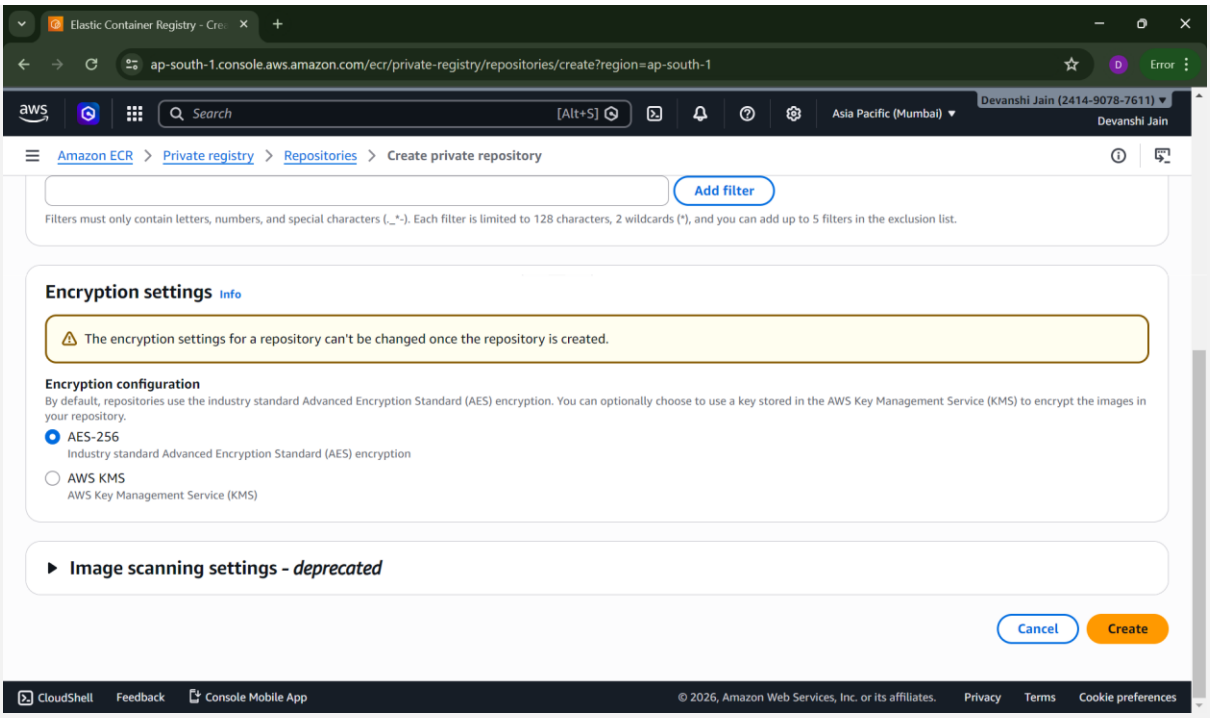
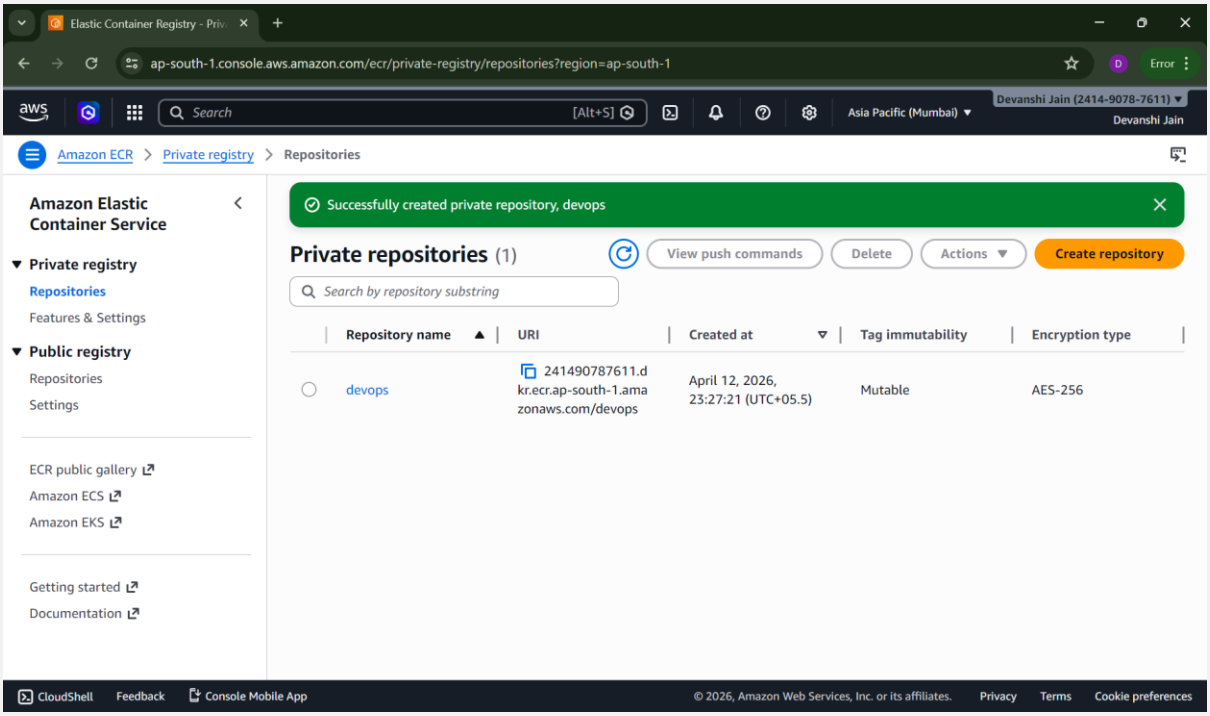
Privacy
Terms
Cookie preferences


Elastic Container Registry - Cre
ap-south-1.console.aws.amazon.com/ecr/private-registry/repositories/create?region=ap-south-1
Search [Alt+S]
Asia Pacific (Mumbai)
Devanshi Jain (2414-9078-7611)
Devanshi Jain
Amazon ECR > Private registry > Repositories > Create private repository
Add filter
Filters must only contain letters, numbers, and special characters (._*-). Each filter is limited to 128 characters, 2 wildcards (*), and you can add up to 5 filters in the exclusion list.
Encryption settings Info
The encryption settings for a repository can't be changed once the repository is created.
Encryption configuration
By default, repositories use the industry standard Advanced Encryption Standard (AES) encryption. You can optionally choose to use a key stored in the AWS Key Management Service (KMS) to encrypt the images in your repository.
AES-256
Industry standard Advanced Encryption Standard (AES) encryption
AWS KMS
AWS Key Management Service (KMS)
Image scanning settings - deprecated
Cancel
Create
CloudShell
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


Elastic Container Registry - Priv
ap-south-1.console.aws.amazon.com/ecr/private-registry/repositories?region=ap-south-1
Search [Alt+S]
Asia Pacific (Mumbai)
Devanshi Jain (2414-9078-7611)
Devanshi Jain
Amazon ECR > Private registry > Repositories
Amazon Elastic Container Service
Private registry
Repositories
Features & Settings
Public registry
Repositories
Settings
ECR public gallery
Amazon ECS
Amazon EKS
Getting started
Documentation
Successfully created private repository, devops
Private repositories (1)
View push commands
Delete
Actions
Create repository
Search by repository substring
Repository name
URI
Created at
Tag immutability
Encryption type
devops
241490787611.dkr.ecr.ap-south-1.amazonaws.com/devops
April 12, 2026, 23:27:21 (UTC+05.5)
Mutable
AES-256
CloudShell
Feedback
Console Mobile App
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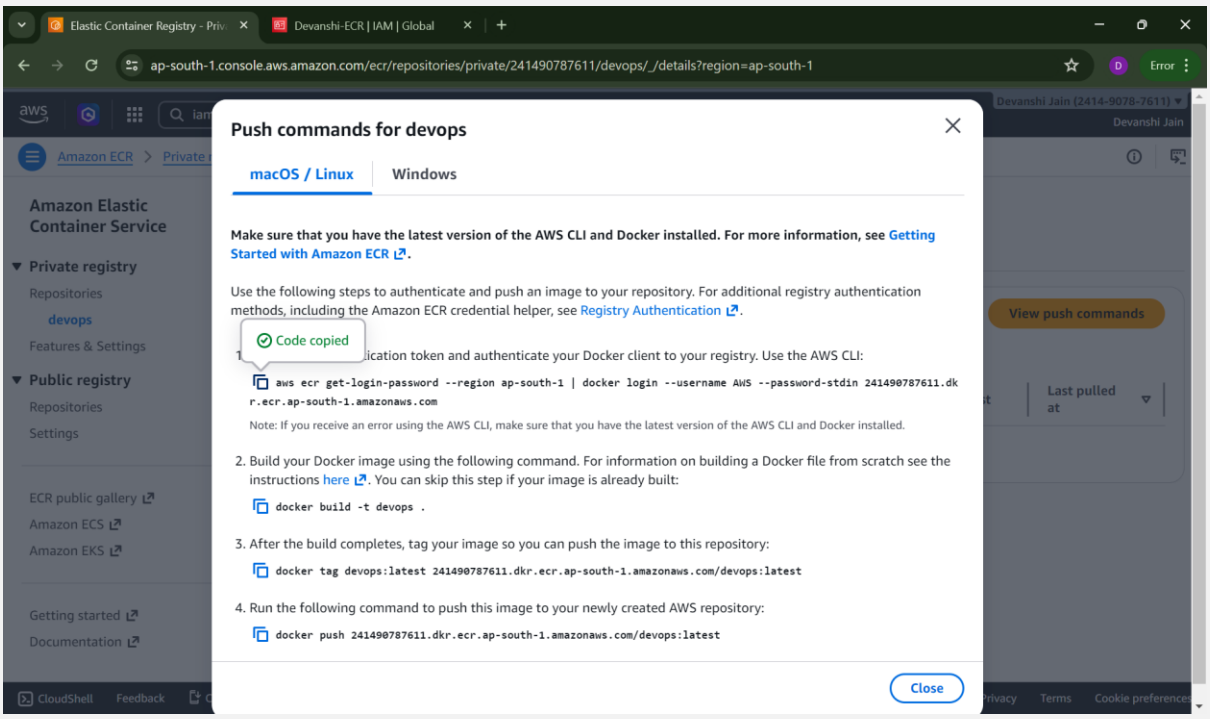
## Task 4: Authenticate Docker to ECR

Login to the ECR registry using **AWS CLI**.

Example:

aws ecr get-login-password --region us-east-1 | docker login --username AWS --password-stdin <account-id>.dkr.ecr.us-east-1.amazonaws.com

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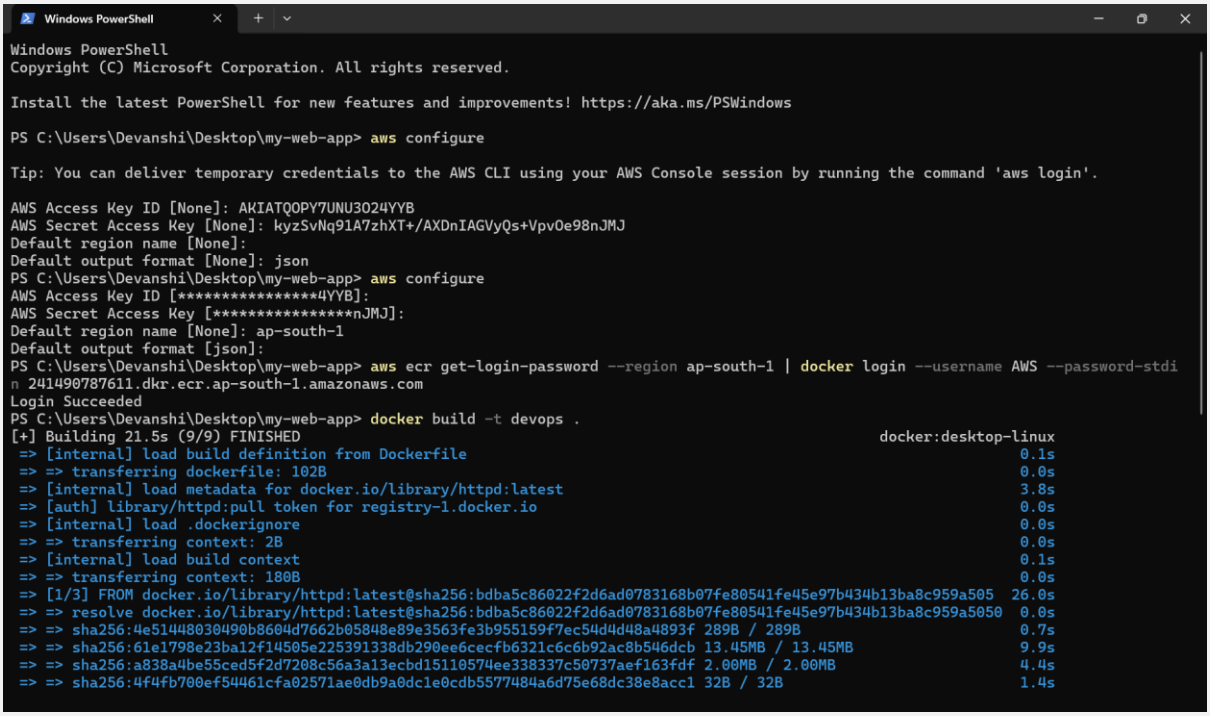
## Task 5: Tag the Docker Image

Tag the image to match the ECR repository.

Example:

```
docker tag ecs-lab-app:latest <account-id>.dkr.ecr.us-east-1.amazonaws.com/ecs-lab-repository:latest
```



```
Windows PowerShell
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Devanshi\Desktop\my-web-app> aws configure

Tip: You can deliver temporary credentials to the AWS CLI using your AWS Console session by running the command 'aws login'.

AWS Access Key ID [None]: AKIATQOPY7UNU3O24YYB
AWS Secret Access Key [None]: kyzSvNq91A7zhXT+/AXDnIAGVyQs+VpvOe98nJMJ
Default region name [None]:
Default output format [None]: json
PS C:\Users\Devanshi\Desktop\my-web-app> aws configure
AWS Access Key ID [****************4YYB]:
AWS Secret Access Key [****************nJMJ]:
Default region name [None]: ap-south-1
Default output format [json]:
PS C:\Users\Devanshi\Desktop\my-web-app> aws ecr get-login-password --region ap-south-1 | docker login --username AWS --password-stdi
n 241490787611.dkr.ecr.ap-south-1.amazonaws.com
Login Succeeded
PS C:\Users\Devanshi\Desktop\my-web-app> docker build -t devops .
[+] Building 21.5s (9/9) FINISHED                                                    docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                  0.1s
 => => transferring dockerfile: 102B                                                                  0.0s
 => [internal] load metadata for docker.io/library/httpd:latest                                       3.8s
 => [auth] library/httpd:pull token for registry-1.docker.io                                          0.0s
 => [internal] load .dockerignore                                                                     0.0s
 => => transferring context: 2B                                                                       0.0s
 => [internal] load build context                                                                     0.1s
 => => transferring context: 180B                                                                     0.0s
 => [1/3] FROM docker.io/library/httpd:latest@sha256:bdba5c86022f2d6ad0783168b07fe80541fe45e97b434b13ba8c959a505  26.0s
 => => resolve docker.io/library/httpd:latest@sha256:bdba5c86022f2d6ad0783168b07fe80541fe45e97b434b13ba8c959a5050  0.0s
 => => sha256:4e51448030490b8604d7662b05848e89e3563fe3b955159f7ec54d4d48a4893f 289B / 289B            0.7s
 => => sha256:61e1798e23ba12f14505e225391338db290ee6cecfb6321c6c6b92ac8b546dcb 13.45MB / 13.45MB      9.9s
 => => sha256:a838a4be55ced5f2d7208c56a3a13ecbd15110574ee338337c50737aef163fdf 2.00MB / 2.00MB        4.4s
 => => sha256:4f4fb700ef54461cfa02571ae0db9a0dc1e0cdb5577484a6d75e68dc38e8acc1 32B / 32B              1.4s
```

---

## Task 6: Push the Image to ECR

Push the image to the repository.

```
docker push <account-id>.dkr.ecr.us-east-1.amazonaws.com/ecs-lab-repository:latest
```

```
Windows PowerShell
=> => sha256:4e51448030490b8604d7662b05848e89e3563fe3b955159f7ec54d4d48a4893f 289B / 289B                    0.7s
=> => sha256:61e1798e23ba12f14505e225391338db290ee6cecfb6321c6c6b92ac8b546dcb 13.45MB / 13.45MB              9.9s
=> => sha256:a838a4be55ced5f2d7208c56a3a13ecbd15110574ee338337c50737aef163fdf 2.00MB / 2.00MB                4.4s
=> => sha256:4f4fb700ef54461cfa02571ae0db9a0dc1e0cdb5577484a6d75e68dc38e8acc1 32B / 32B                      1.4s
=> => sha256:8261ede62f8c1c85c3e4b2ae3a9ae5a586b84fcc61b60c110ee65c9c523b49c7 146B / 146B                    0.7s
=> => sha256:5435b2dcdf5cb7faa0d5b1d4d54be2c72a776fab9a605336f5067d6e9ecb5976 29.78MB / 29.78MB             24.4s
=> => extracting sha256:5435b2dcdf5cb7faa0d5b1d4d54be2c72a776fab9a605336f5067d6e9ecb5976                     1.0s
=> => extracting sha256:8261ede62f8c1c85c3e4b2ae3a9ae5a586b84fcc61b60c110ee65c9c523b49c7                     0.0s
=> => extracting sha256:4f4fb700ef54461cfa02571ae0db9a0dc1e0cdb5577484a6d75e68dc38e8acc1                     0.0s
=> => extracting sha256:a838a4be55ced5f2d7208c56a3a13ecbd15110574ee338337c50737aef163fdf                     0.1s
=> => extracting sha256:61e1798e23ba12f14505e225391338db290ee6cecfb6321c6c6b92ac8b546dcb                     0.3s
=> => extracting sha256:4e51448030490b8604d7662b05848e89e3563fe3b955159f7ec54d4d48a4893f                     0.0s
=> [2/3] WORKDIR /usr/local/apache2/htdocs/                                                                  0.2s
=> [3/3] COPY index.html .                                                                                   0.0s
=> exporting to image                                                                                        0.3s
=> => exporting layers                                                                                       0.1s
=> => exporting manifest sha256:c2dc1f16339b60f1832a011664244a2ab1a9eecd0b0049eeeebbc0d5d1c6c9f6             0.0s
=> => exporting config sha256:9391ab1201a9c36efde3b1d32177669050c1d33e9028d722cbfd84b8b58e2c62               0.0s
=> => exporting attestation manifest sha256:ce8114b93b18d8ca0d54102c61d3b4aad98183ed7bf6329a1a3f93b79b127930 0.0s
=> => exporting manifest list sha256:91bca42beb9eee26668354811453ca671eaef290ad2ab61da5d1fd13877c0022        0.0s
=> => naming to docker.io/library/devops:latest                                                              0.0s
=> => unpacking to docker.io/library/devops:latest                                                           0.1s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/ssvwk5zhdmujq79evnlxuz37w
PS C:\Users\Devanshi\Desktop\my-web-app> docker tag devops:latest 241490787611.dkr.ecr.ap-south-1.amazonaws.com/devops:latest
PS C:\Users\Devanshi\Desktop\my-web-app> docker push 241490787611.dkr.ecr.ap-south-1.amazonaws.com/devops:latest
The push refers to repository [241490787611.dkr.ecr.ap-south-1.amazonaws.com/devops]
271b32e4df8a: Pushed
4f4fb700ef54: Pushed
5435b2dcdf5c: Pushed
8261ede62f8c: Pushed
4e5144803049: Pushed
a838a4be55ce: Pushed
61e1798e23ba: Pushed
82dd8462d38b: Pushed
latest: digest: sha256:91bca42beb9eee26668354811453ca671eaef290ad2ab61da5d1fd13877c0022 size: 856
PS C:\Users\Devanshi\Desktop\my-web-app>
```

Verify the image in the **ECR console**.

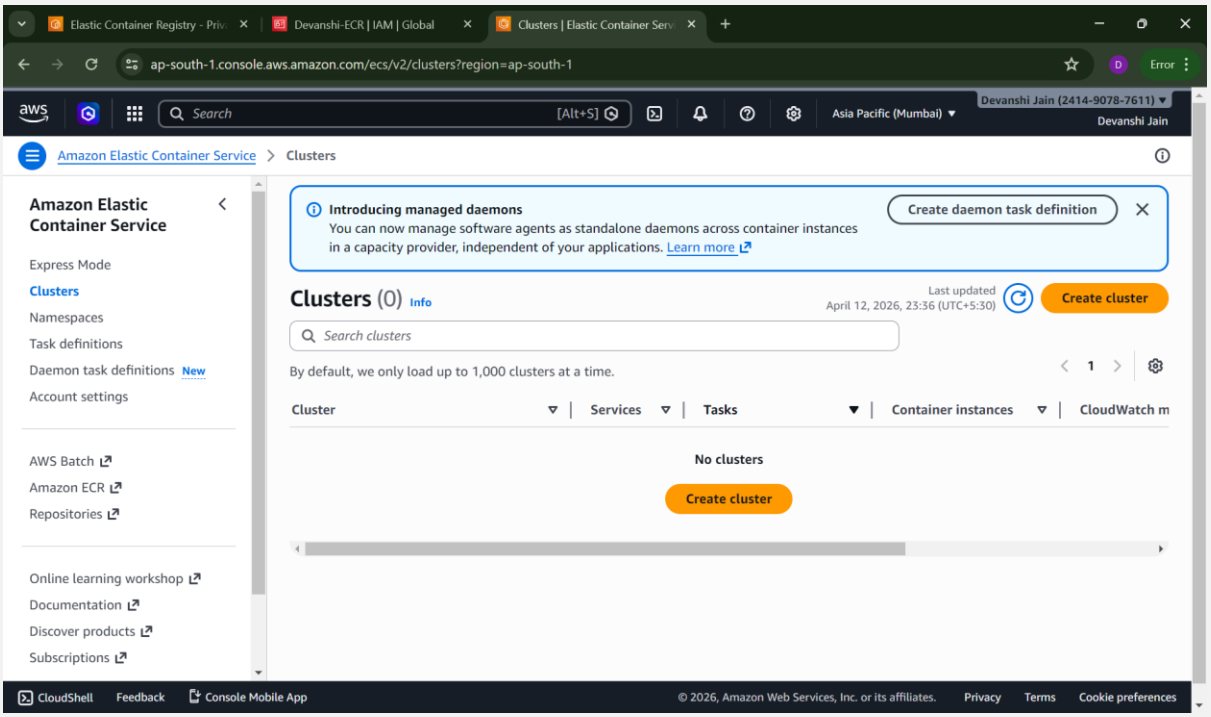
## Task 7: Create ECS Cluster

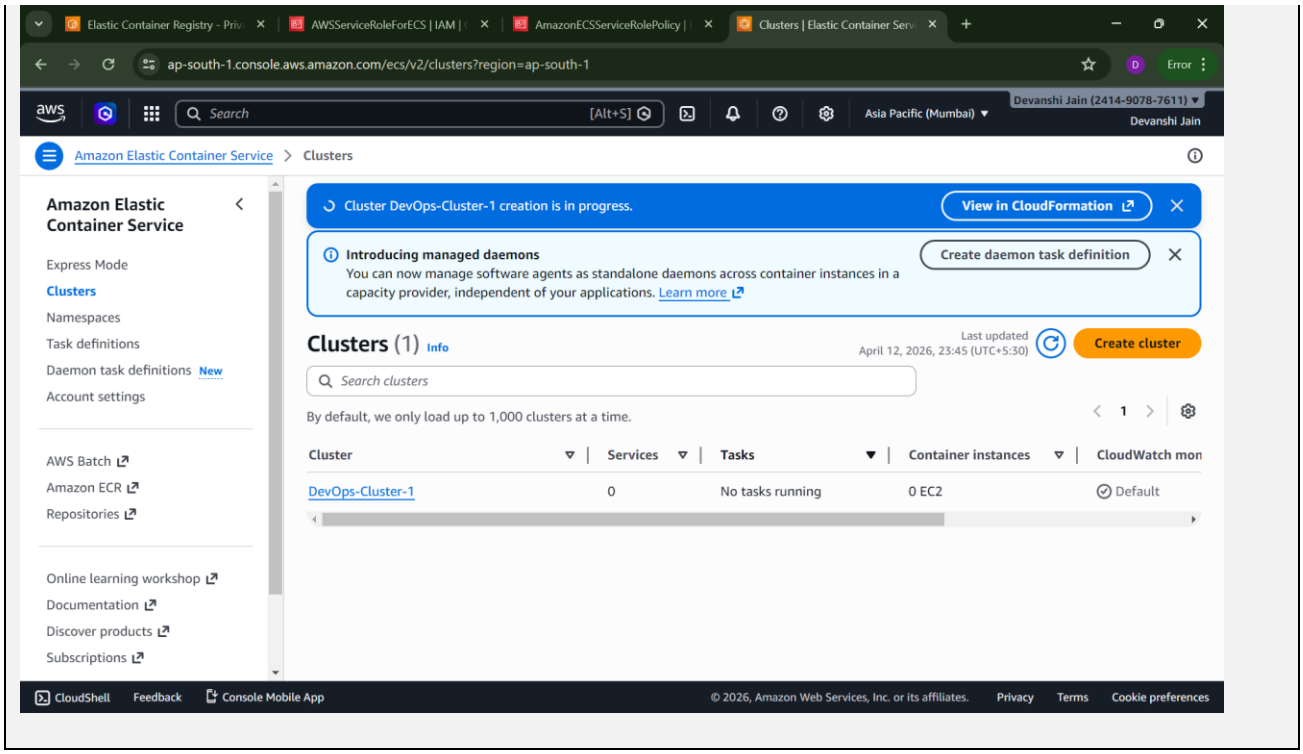
Navigate to **Amazon ECS**.

Steps:

1. Click **Create Cluster**
2. Select **Fargate**
3. Provide cluster name:

ecs-lab-cluster






---

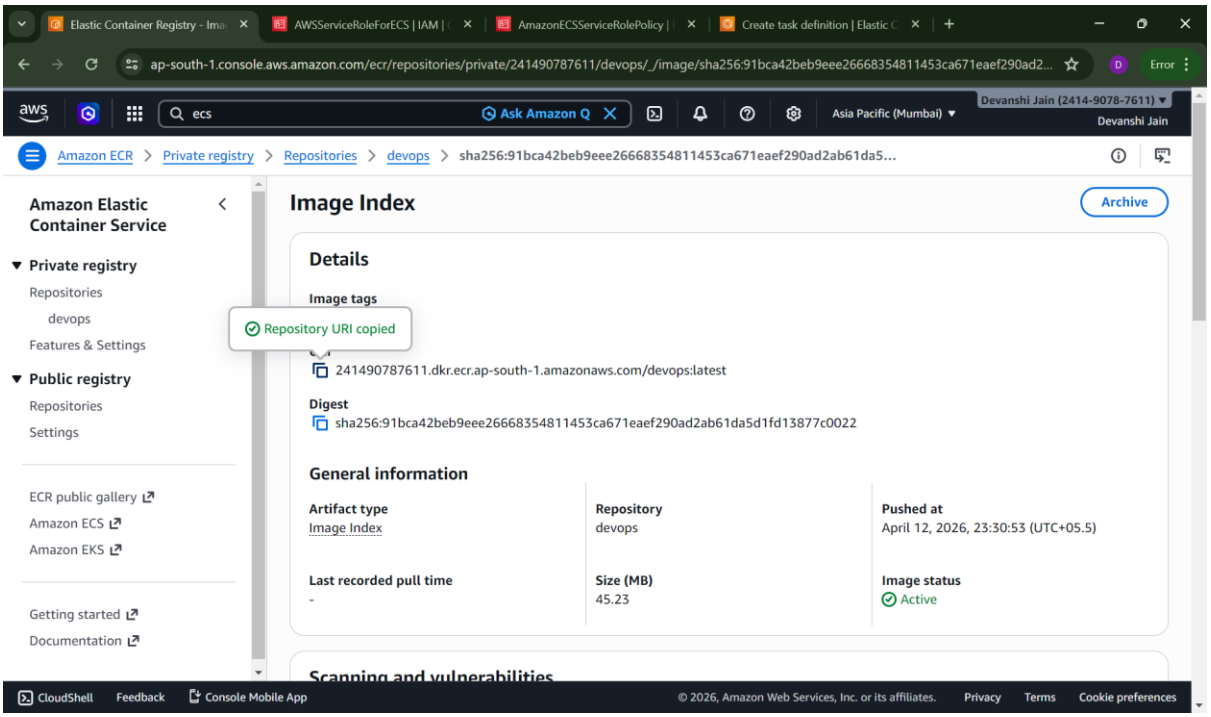
## Task 8: Create Task Definition

Create a task definition.

Settings:

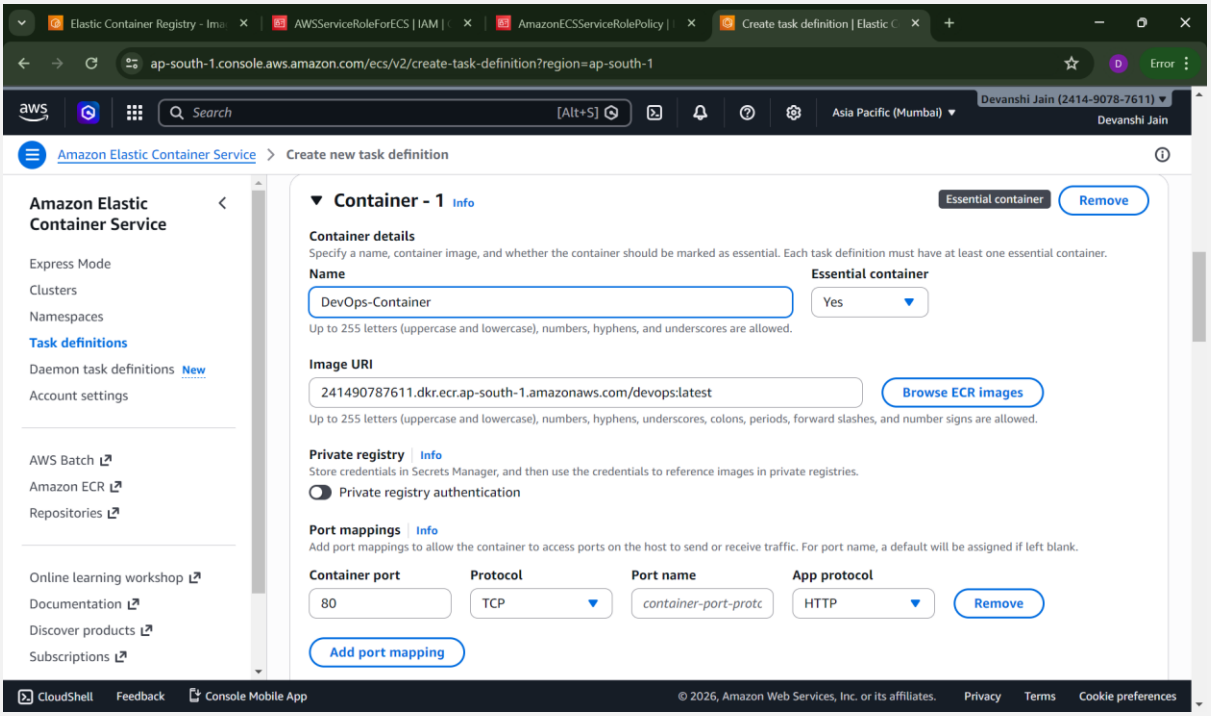
Container image:

```
<account-id>.dkr.ecr.us-east-1.amazonaws.com/ecs-lab-repository:latest
```




Port mapping:

80

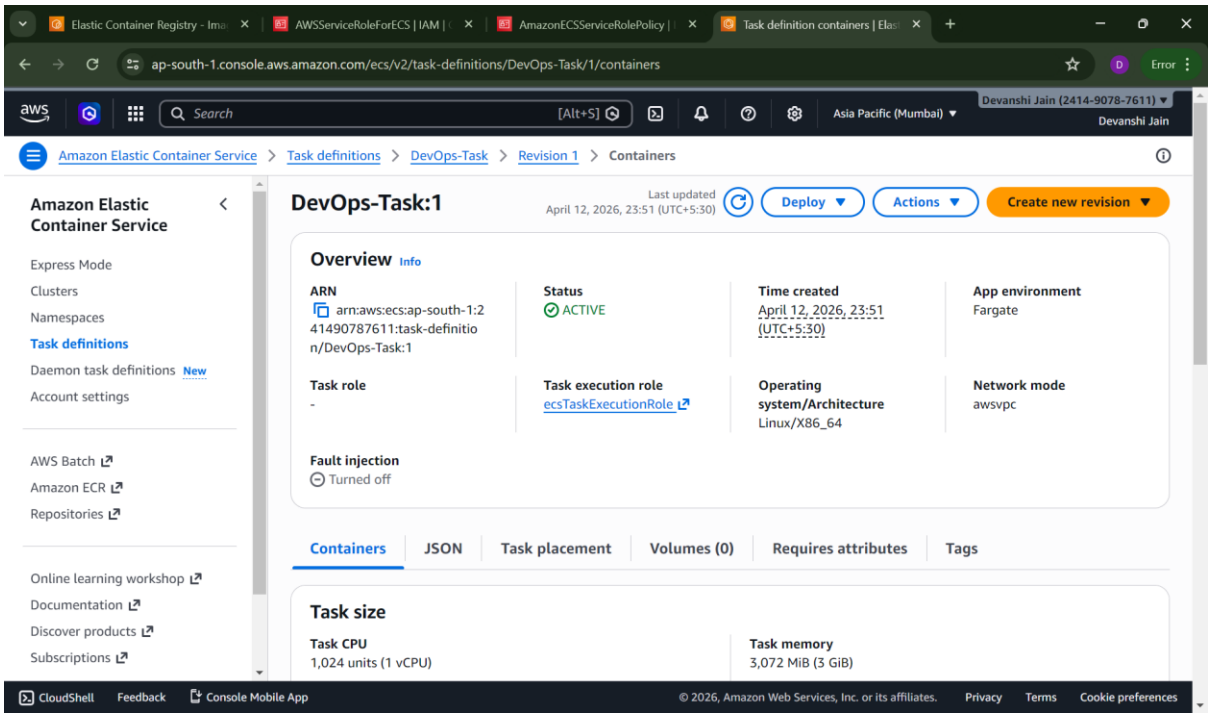


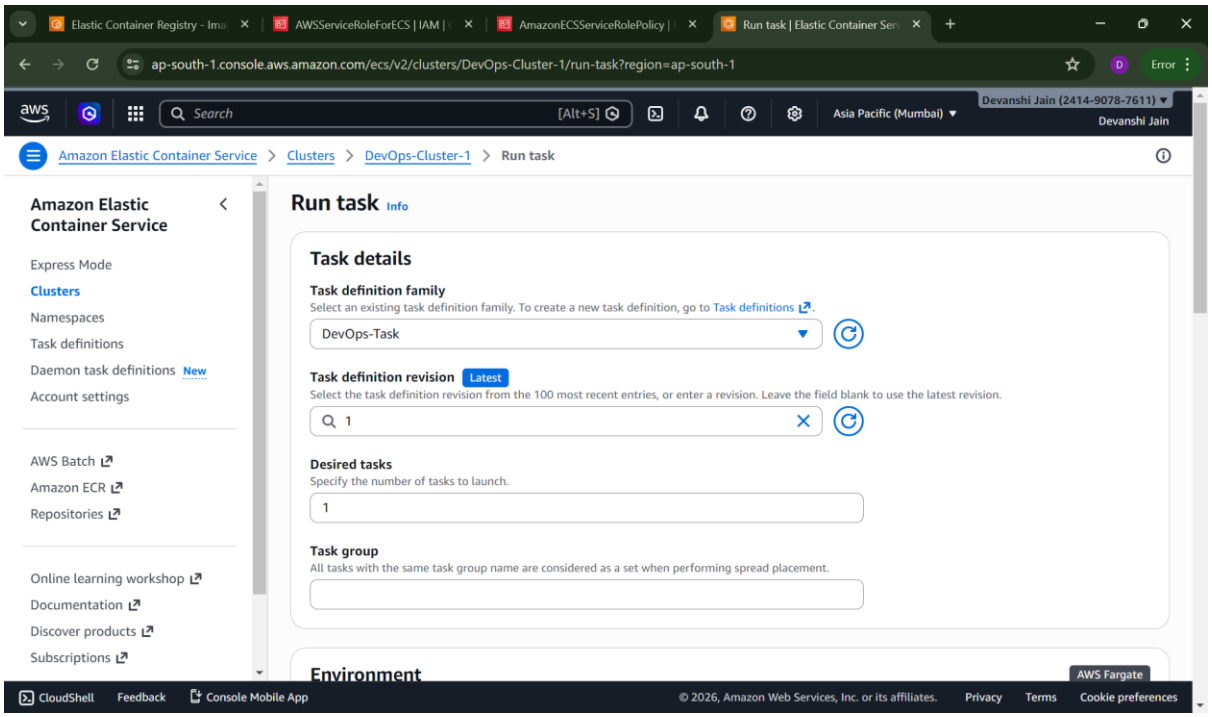
## Task 9: Run ECS Service

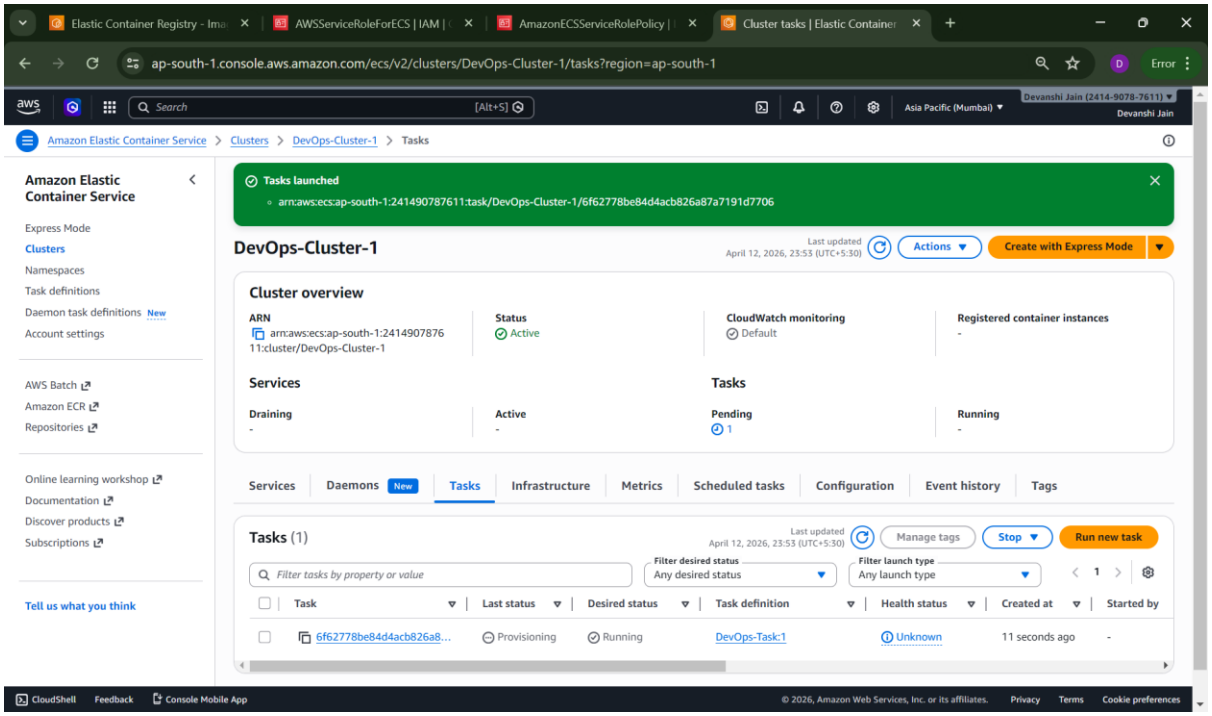
Deploy the container.

Steps:

1. Create service
2. Select **Fargate**
3. Desired tasks:
4. Enable public IP






---

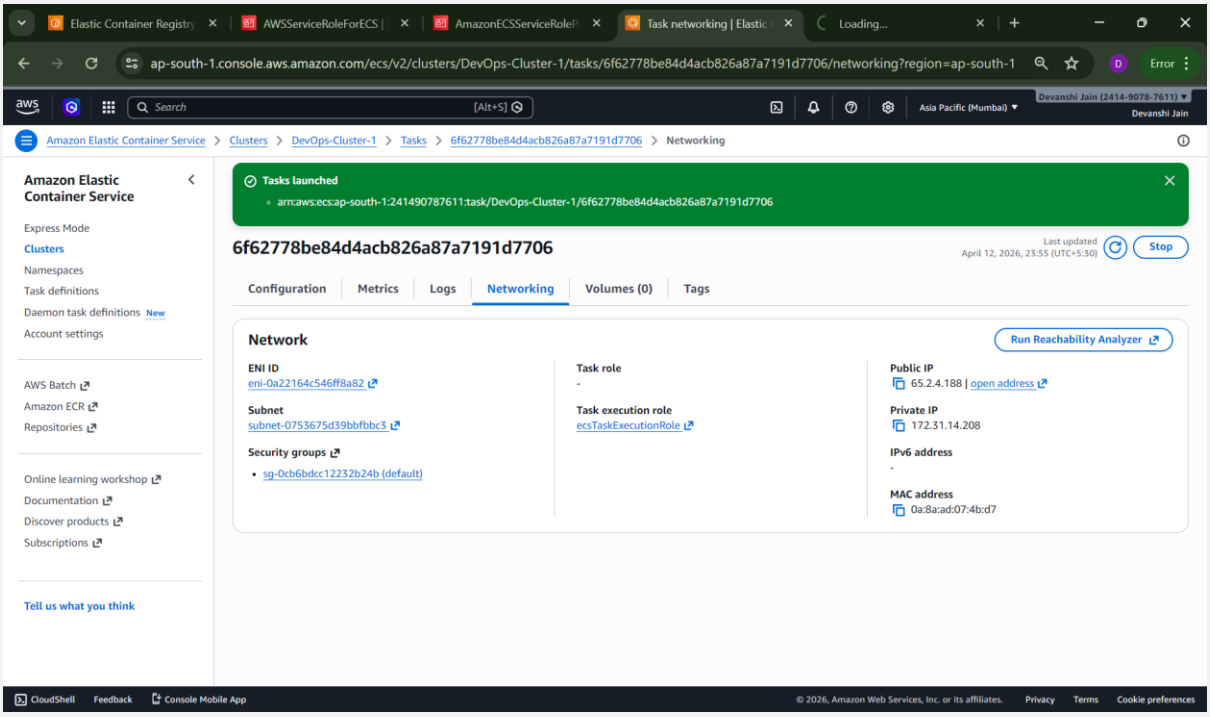
## Task 10: Verify Deployment

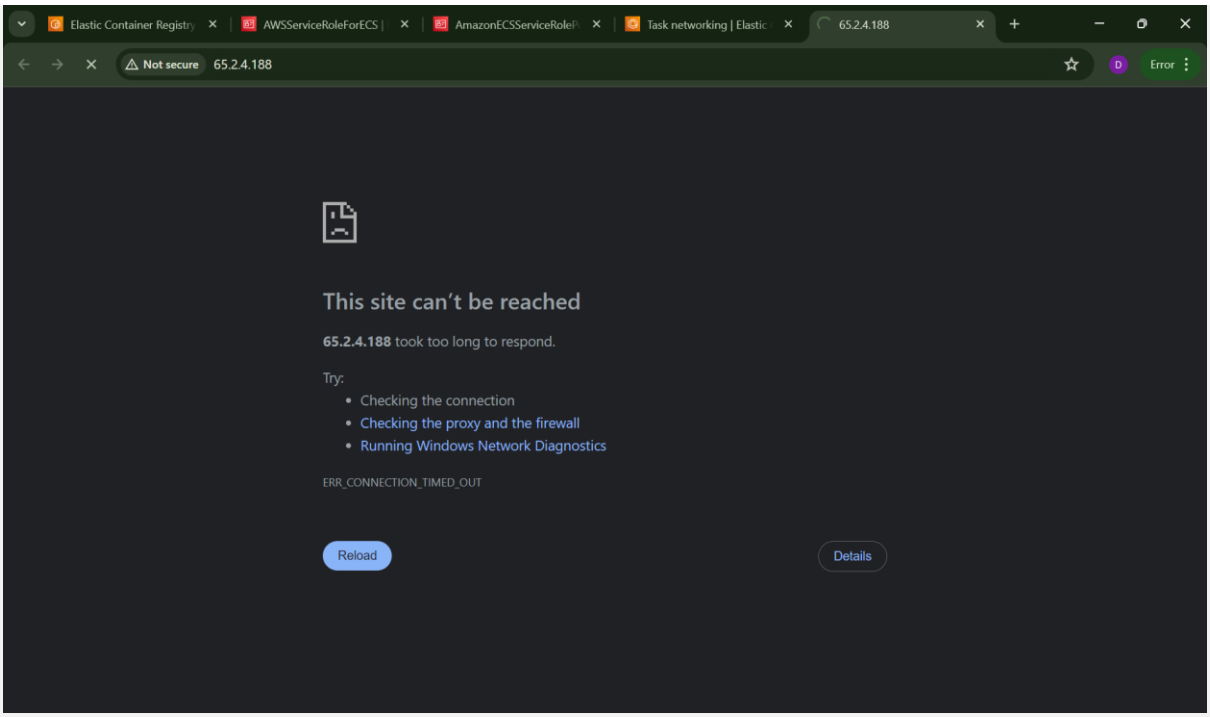
After deployment:

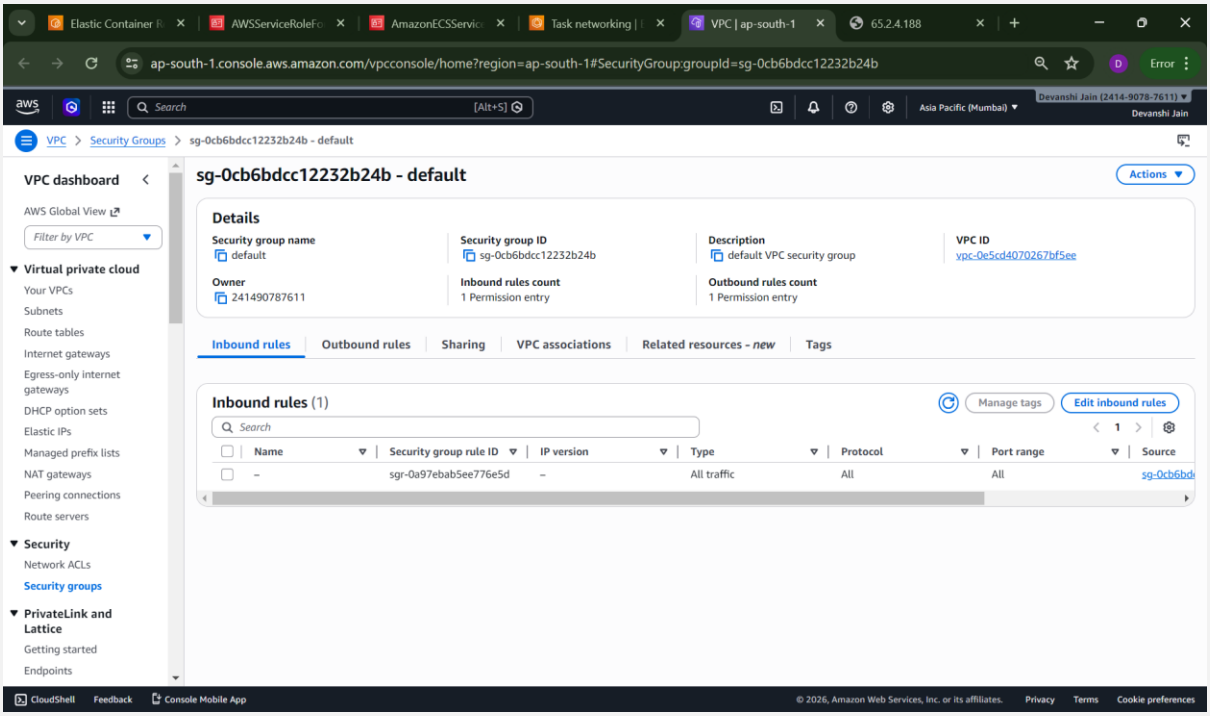
1. Open ECS service
2. Copy the **public IP or load balancer URL**
3. Open in browser

Expected output:

Hello from AWS ECS Lab




Elastic Container Registry
AWSServiceRoleForECS
AmazonECSServiceRoleF
Task networking | Elastic
65.2.4.188
Not secure 65.2.4.188
Error
This site can't be reached
65.2.4.188 took too long to respond.
Try:
Checking the connection
Checking the proxy and the firewall
Running Windows Network Diagnostics
ERR_CONNECTION_TIMED_OUT
Reload
Details


VPC | ap-south-1
ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#SecurityGroup:groupId=sg-0cb6bdcc12232b24b
Asia Pacific (Mumbai)
Devanshi Jain
VPC > Security Groups > sg-0cb6bdcc12232b24b - default
VPC dashboard
AWS Global View
Filter by VPC
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
Security
Network ACLs
Security groups
PrivateLink and Lattice
Getting started
Endpoints
sg-0cb6bdcc12232b24b - default
Actions
Details
Security group name
default
Security group ID
sg-0cb6bdcc12232b24b
Description
default VPC security group
VPC ID
vpc-0e5cd4070267bf5ee
Owner
241490787611
Inbound rules count
1 Permission entry
Outbound rules count
1 Permission entry
Inbound rules
Outbound rules
Sharing
VPC associations
Related resources - new
Tags
Inbound rules (1)
Manage tags
Edit inbound rules
Name
Security group rule ID
IP version
Type
Protocol
Port range
Source
sgr-0a97ebab5ee776e5d
All traffic
All
All
CloudShell
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
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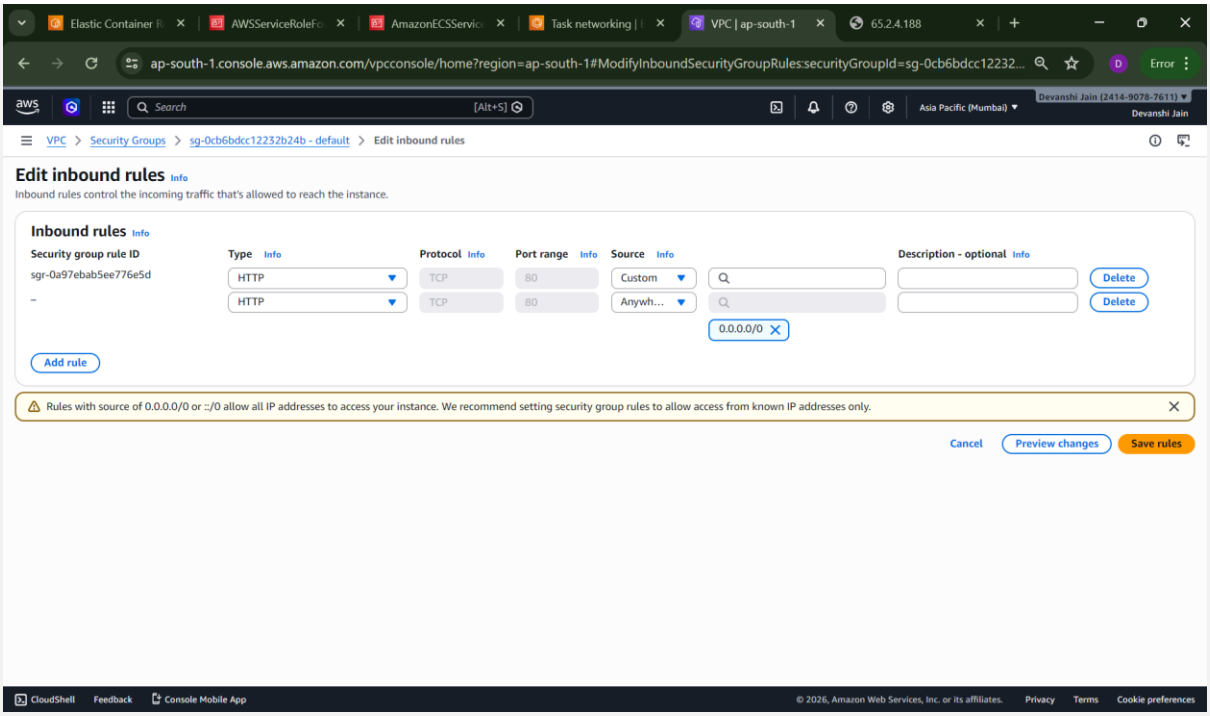
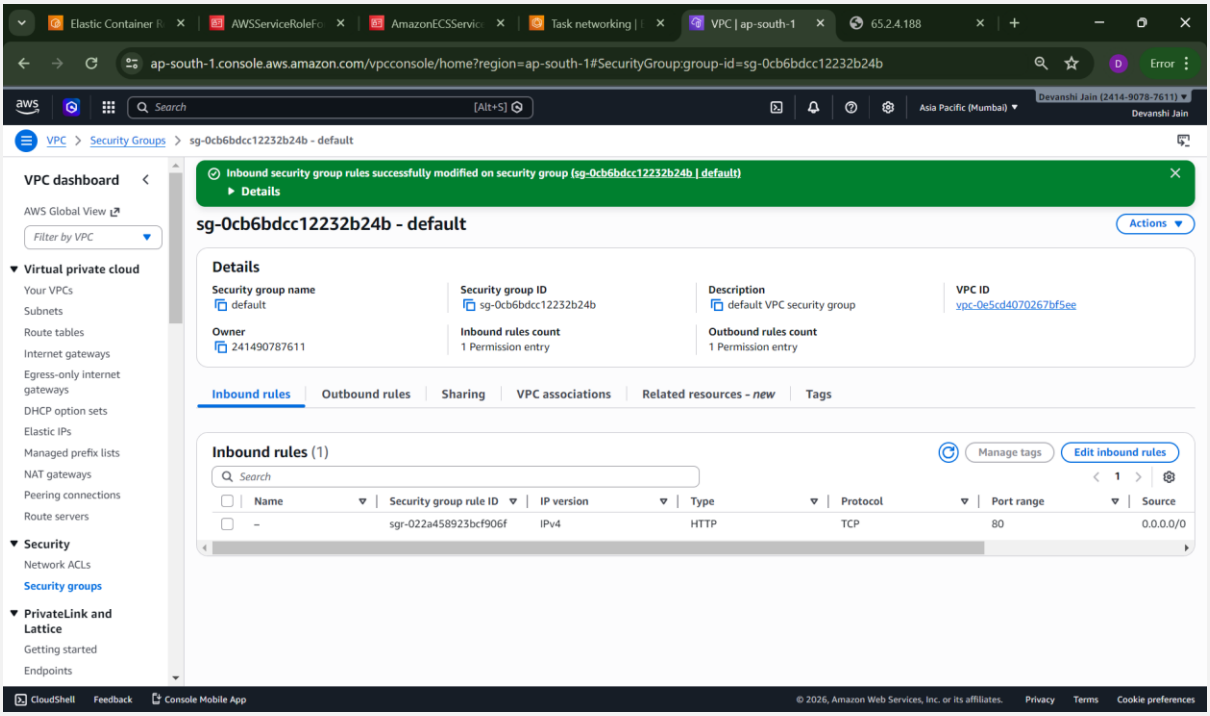

Elastic Container
AWSServiceRoleFo
AmazonECSServic
Task networking |
VPC | ap-south-1
65.2.4.188
ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#ModifyInboundSecurityGroupRules:securityGroupId=sg-0cb6bdcc12232...
Error
Search
[Alt+S]
Asia Pacific (Mumbai)
Devanshi Jain (2414-9078-7611)
Devanshi Jain
VPC > Security Groups > sg-0cb6bdcc12232b24b - default > Edit inbound rules
Edit inbound rules Info
Inbound rules control the incoming traffic that's allowed to reach the instance.
Inbound rules Info
Security group rule ID
Type Info
Protocol Info
Port range Info
Source Info
Description - optional Info
sgr-0a97ebab5ee776e5d
HTTP
TCP
80
Custom
Delete
-
HTTP
TCP
80
Anywh...
0.0.0.0/0
Delete
Add rule
Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.
Cancel
Preview changes
Save rules
CloudShell
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


Elastic Container
AWSServiceRoleFo
AmazonECSServic
Task networking |
VPC | ap-south-1
65.2.4.188
ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#SecurityGroup:group-id=sg-0cb6bdcc12232b24b
Error
Search
[Alt+S]
Asia Pacific (Mumbai)
Devanshi Jain (2414-9078-7611)
Devanshi Jain
VPC > Security Groups > sg-0cb6bdcc12232b24b - default
VPC dashboard
AWS Global View
Filter by VPC
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
Security
Network ACLs
Security groups
PrivateLink and Lattice
Getting started
Endpoints
Inbound security group rules successfully modified on security group (sg-0cb6bdcc12232b24b | default)
Details
sg-0cb6bdcc12232b24b - default
Actions
Details
Security group name
default
Security group ID
sg-0cb6bdcc12232b24b
Description
default VPC security group
VPC ID
vpc-0e5cd4070267bf5ee
Owner
241490787611
Inbound rules count
1 Permission entry
Outbound rules count
1 Permission entry
Inbound rules
Outbound rules
Sharing
VPC associations
Related resources - new
Tags
Inbound rules (1)
Manage tags
Edit inbound rules
Search
Name
Security group rule ID
IP version
Type
Protocol
Port range
Source
-
sgr-022a458923bcf906f
IPv4
HTTP
TCP
80
0.0.0.0/0
CloudShell
Feedback
Console Mobile App
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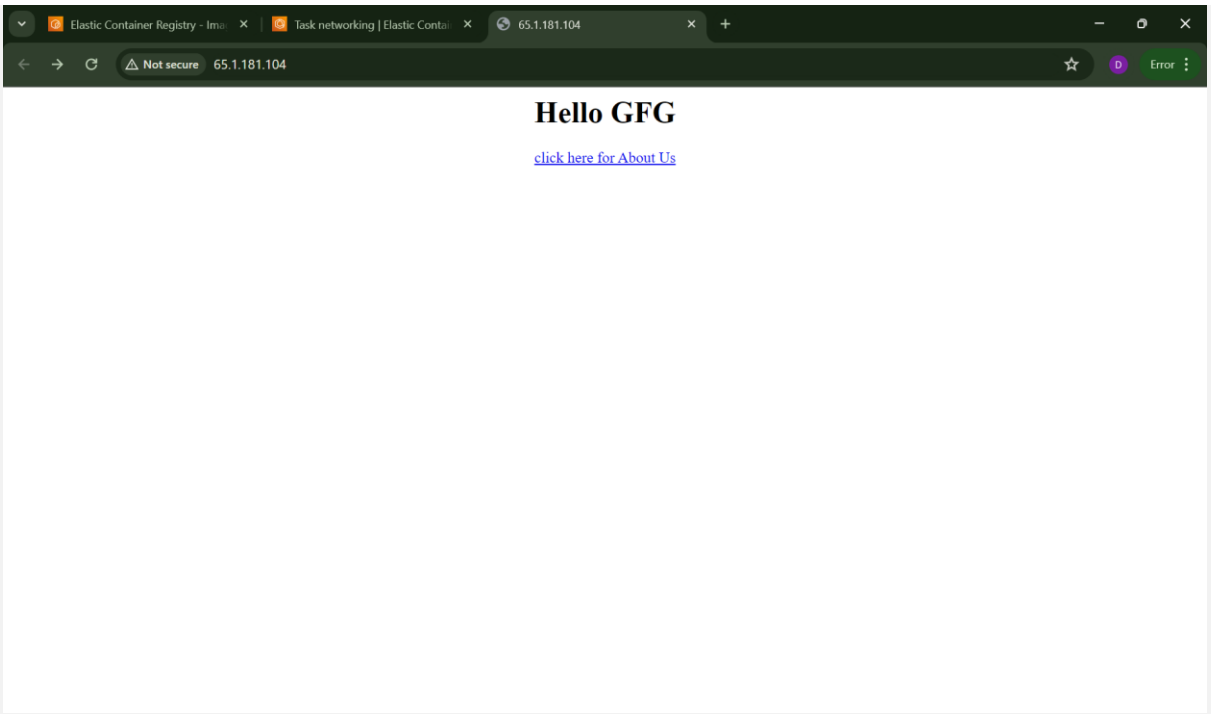
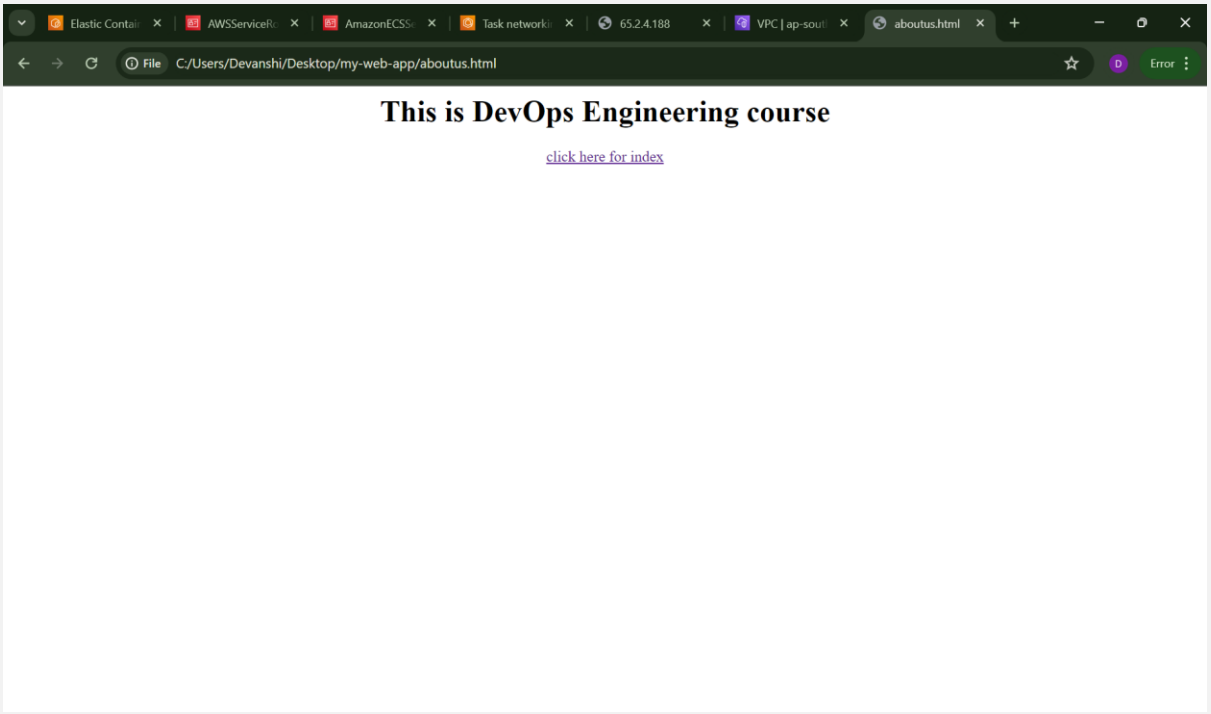
Privacy
Terms
Cookie preferences

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## 5. Expected Outcome

After completing this lab:

- Docker image is stored in **Amazon ECR**
- Container is running on **Amazon ECS**
- Application is accessible through a browser